Structure vs Sequence Report: Mycgen - Mycoplasmoides_genitalium_G37

This report compares the detection capabilities of:

- **Sequence methods**: blast
- **Structural methods**: tmalign, reseek, foldseek

0. Theoretical introduction

I chose *reseek* and *foldseek* for comparison because they are created on same idea but *reseek* is more complex model.

Those models add additional features (alphabets) with fixed dimensions (symbols) which are used to describe certain property of folded protein. As result we obtain for every residue tuple of symbols and we can compare every feature with corresponding datasets using standard local and global alignments algorithms translating the NP-hard problem of 3D spatial superposition into a 1D sequence alignment problem, which can be solved in polynomial time $\mathcal{O}(N^2L^2)$ using dynamic programming.

Model complexity:

- foldseek:
  - 1 additional alphabet with 20 symbols ($\underbrace{20}*{\text{20 amino acids}} \times \underbrace{20}*{20 symbols} = 320$ possible states)
- reseek:
  - 14 additional alphabets with 16 symbols each ($\underbrace{20}_{\text{20 amino acids}} \times \underbrace{16^{14}} \approx 10^{18}$ possible states)

Bibliography

I mostly based on Robert Edgar's seminar - Link

1. Metadata analysis

Description

This table summarizes the performance, runtime, and total hit count across all executed methods:

method	runtime_seconds	total_queries	queries_with_hits	queries_0_hits
blast	1.5	433	180	253
tmalign	2235.55	433	433	0
reseek	3.84	433	429	4
foldseek	7.5	433	170	263

We can see that only *tmalign* finds hits for all queries, that is because it compares all structures so it always finds some result regardless of biological relevance. Reseek finds much more results than *foldseek*, that is

because *reseek* alignment score score consists of more features (alphabets).

Methodology

Here we filter results only by E-score (probability of getting hit by randomly creating sequence), to check sensitivity.

2. Sequential vs Structure comparison

Description

Structural Method	Found by Both	False Positives (Seq only)	False Negatives (Struct only)
tmalign	180	0	211
reseek	178	2	63
foldseek	153	27	14

False negatives represent sequences missed by sequential analysis (BLAST) but successfully mapped by structural methods. We assume that those structures are homologs. False positive are those sequences that are found by *blast* but are not found by structural methods. we can see that most effective method is *tmalign*, but is much more complex.

We should also notice that ($253-211=42$) can be true negatives because they were not captured by *tmaling* as well as *blast*. It support thesis that structures are more conserved than sequences.

Methodology

We filter results by TMscore to obtain only homologous results. TMscore in *reseek* and *foldseek* is estimated.

False Negatives (Top 10 by score)

Method: TMALIGN

query	target	score	evalue
WP_234943728	WP_020862785	0.974	0
WP_261156652	WP_020862785	0.964	0
WP_009886004	WP_020862785	0.9544	0
WP_009885614	WP_020863090	0.8994	0
WP_014894048	WP_013729896	0.8948	0
WP_010869411	WP_020862584	0.8701	0
WP_009885968	WP_020862755	0.8637	0
WP_009885899	WP_013729497	0.857	0
WP_009885821	WP_008362910	0.8548	0

query	target	score	evalue
WP_009885894	WP_020862476	0.8525	0

Method: RESEEK

query	target	score	evalue
WP_009885894	WP_020862476	0.9838	1.38e-07
WP_010869332	WP_020863005	0.9784	4.09e-07
WP_009885807	WP_020862878	0.9768	5.34e-07
WP_009885568	WP_020863082	0.9729	9.62e-07
WP_009885968	WP_020862755	0.9686	1.67e-06
WP_009885829	WP_020862637	0.9685	1.7e-06
WP_009885621	WP_017698247	0.9665	2.14e-06
WP_010869411	WP_020862584	0.9647	2.61e-06
WP_009885989	WP_020863013	0.963	3.14e-06
WP_010869439	WP_020862921	0.9623	3.37e-06

Method: FOLDSEEK

query	target	score	evalue
WP_010869411	WP_020862584	0.8782	5.081e-11
WP_009885634	WP_013729566	0.8278	7.288e-11
WP_009885894	WP_020862941	0.8209	3.018e-18
WP_009885786	WP_020862730	0.8026	2.137e-15
WP_009885967	WP_020862509	0.802	3.171e-16
WP_010869439	WP_020862509	0.7346	4.062e-16
WP_010869456	WP_020862476	0.731	1.759e-12
WP_009885813	WP_020863079	0.7243	6.365e-11
WP_010869327	WP_013729180	0.7154	6.567e-11
WP_010869332	WP_020863005	0.6882	2.387e-18

Visual comparison

I choose two pairs of structures:

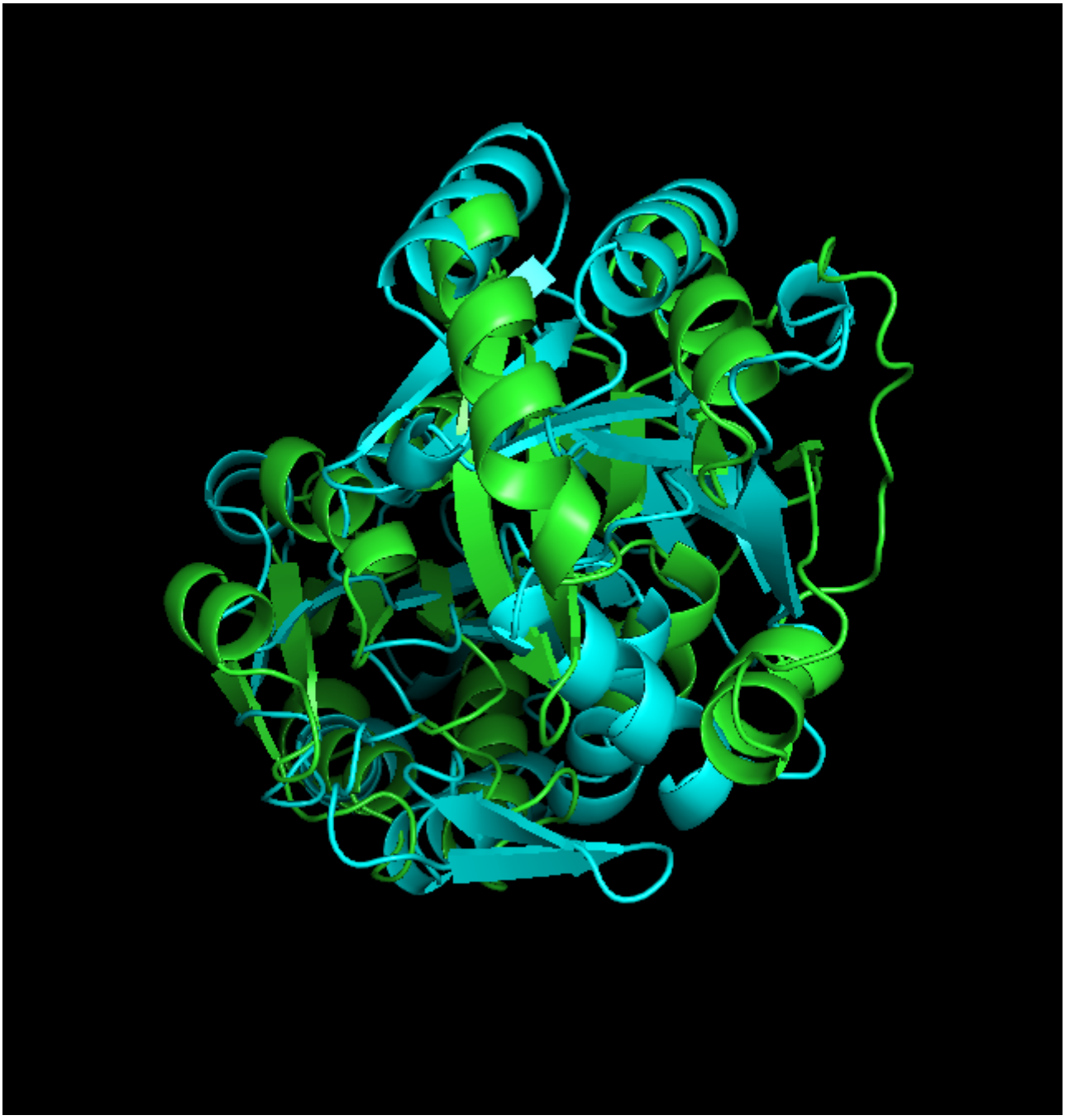
- WP_009885894 vs WP_020862476 (combination appears in *tmalign* and *reseek*)

- WP_010869411 vs WP_020862584 (combination appears in all three methods)

In both cases i tried to find perspective where both sequences differ.

WP_009885894 vs WP_020862476

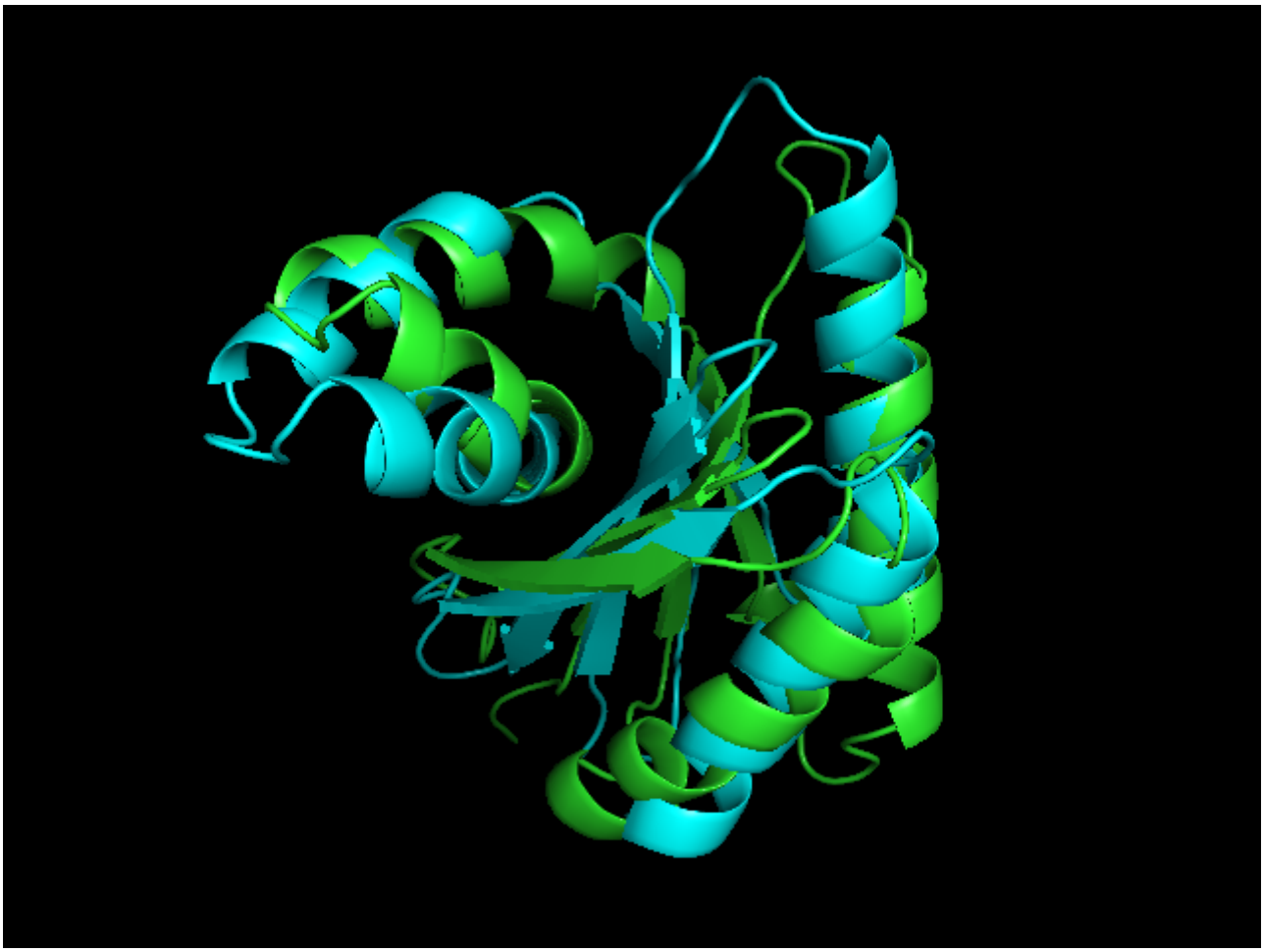
- **Cyan**: Query sequence from *Mycoplasmoides genitalium* G37
- **Green**: Target sequence from *CP014940*



Here helixes are little displaced.

WP_010869411 vs WP_020862584

- **Cyan**: Query sequence from *Mycoplasmoides genitalium* G37
- **Green**: Target sequence from *CP014940*



Both structures share a nearly identical core fold, with secondary structures perfectly aligned. Minor deviations are visible mostly on the flexible loop regions.